# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** From a solid cylinder of height 48 cm and base radius 14 cm, a conical cavity of the same height and the same base radius is hollowed out. The total surface area of the remaining solid is: 1

**(a)** 7040 cm² **(b)** 5456 cm² **(c)** 6424 cm² **(d)** 7656 cm²

**2.** A wooden article is made by scooping out a hemisphere of radius 7 cm from each end of a solid cylinder of height 15 cm and base radius 7 cm. The total surface area of the article is: 1

**(a)** 660 cm² **(b)** 1584 cm² **(c)** 968 cm² **(d)** 1276 cm²

**3.** A solid metallic cylinder of base radius 6 cm and height 24 cm is melted and recast into solid spheres of radius 2 cm. The number of spheres obtained is: 1

**(a)** 162 **(b)** 81 **(c)** 79 **(d)** 87

**4.** A solid metallic cone of base radius 3 cm and height 12 cm is melted and recast into a solid sphere. The radius of the sphere is: 1

**(a)** 3 cm **(b)** 4 cm **(c)** 6 cm **(d)** 2 cm

**5.** A toy is a cone of base radius 35 cm and height 12 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

**(a)** 15620 cm² **(b)** 11770 cm² **(c)** 9020 cm² **(d)** 4070 cm²

**6.** A capsule is a cylinder of radius 7 cm and length 20 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

**(a)** 1496 cm² **(b)** 1188 cm² **(c)** 2992 cm² **(d)** 1804 cm²

**7.** A solid is a cube of edge 7 cm surmounted by a hemisphere of diameter 7 cm (the hemisphere rests on a face of the cube). The total surface area of the solid is: 1

**(a)** 294 cm² **(b)** 255.5 cm² **(c)** 332.5 cm² **(d)** 371 cm²

**8.** From a solid cylinder of height 28 cm and base radius 21 cm, a conical cavity of the same height and the same base radius is hollowed out. The total surface area of the remaining solid is: 1

**(a)** 7392 cm² **(b)** 8778 cm² **(c)** 6468 cm² **(d)** 6006 cm²

**9.** A wooden article is made by scooping out a hemisphere of radius 7 cm from each end of a solid cylinder of height 20 cm and base radius 7 cm. The total surface area of the article is: 1

**(a)** 880 cm² **(b)** 1496 cm² **(c)** 1804 cm² **(d)** 1188 cm²

**10.** A solid metallic cylinder of base radius 3 cm and height 32 cm is melted and recast into solid spheres of radius 2 cm. The number of spheres obtained is: 1

**(a)** 25 **(b)** 54 **(c)** 33 **(d)** 27

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** A solid metal cuboid 24 cm by 12 cm by 15 cm is melted and recast into cubes of edge 3 cm. Find the number of cubes obtained. 2

**12.** A solid sphere of radius 6 cm is completely dropped into water in a cylindrical vessel of base radius 4 cm, and no water spills. Find the rise in the water level. 2

**13.** A solid metallic sphere of radius 3 cm is melted and drawn into a wire of circular cross-section of radius 1 cm. Find the length of the wire. 2

**14.** A solid metal cuboid 24 cm by 18 cm by 12 cm is melted and recast into cubes of edge 6 cm. Find the number of cubes obtained. 2

## SECTION — C (Short Answer Type) 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A toy is a cone of base radius 10.5 cm and height 14 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

**16.** A capsule is a cylinder of radius 21 cm and length 15 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

**17.** From a solid cylinder of height 15 cm and base radius 7 cm, a conical cavity of the same height and base radius is hollowed out. Find the volume of the remaining solid. 3

**OR**

A toy is a cone of base radius 10.5 cm and height 28 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

**18.** A capsule is a cylinder of radius 21 cm and length 30 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

**OR**

From a solid cylinder of height 9 cm and base radius 14 cm, a conical cavity of the same height and base radius is hollowed out. Find the volume of the remaining solid. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Water from a full cylindrical tank of radius 7 cm and height 15 cm is poured into an empty rectangular tank of base 21 cm by 11 cm. Find the height of water in the rectangular tank. 5

**OR**

A conical vessel of radius 6 cm and height 15 cm is full of water. The water is poured into a cylindrical vessel of base radius 5 cm. Find the height of the water in the cylindrical vessel. 5

**20.** Water from a full cylindrical tank of radius 7 cm and height 24 cm is poured into an empty rectangular tank of base 33 cm by 14 cm. Find the height of water in the rectangular tank. 5

**OR**

A conical vessel of radius 6 cm and height 10 cm is full of water. The water is poured into a cylindrical vessel of base radius 5 cm. Find the height of the water in the cylindrical vessel. 5

## END OF QUESTION PAPER Total: 40 Marks